

Benjamin Chou

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Professional Summary

Final-year Purdue ECE PhD student researching audio, music, and multimodal AI/ML: personalized speech enhancement, note-level music separation, music error detection with Transformers, and training-free Multimodal LLMs. Proficient in Audio DSP, Python, PyTorch, TensorFlow, and HPC. Organized three AI4Music workshops.

Education

Ph.D. in Electrical & Computer Engineering | *Purdue University*, West Lafayette, IN 2023 - Present

- GPA: 3.8/4.0
- Expected graduation: Sep 2027 (tentative)

B.Sc. in Electrical Engineering | *National Cheng Kung University*, Tainan, Taiwan 2020 - 2023

- GPA: 4.0/4.3

Experience

Audio Applied Research Scientist Intern | *Shure*, Niles, IL Jul 2026 - Present

- Researching personalized speech enhancement and target speech extraction for multi-speaker conferencing systems.
- Investigating small, real-time-capable generative models for conferencing audio (**flow matching**, **MeanFlow**).

PhD Resident | *Google X*, Mountain View, CA Apr 2026 - Jul 2026

- Developed GraphRAG and knowledge graphs for business applications, planning, reasoning, execution, and agent synthesis.
- Built coding and code-review agents (**LangChain**, **Vertex AI**, **RAG**, **agent skills**, **loop engineering**, **Google Cloud Platform (GCP)**, **GitHub CLI**, **Linear**).

Applied Research Scientist Intern | *Reality Defender*, Manhattan, New York May 2025 - Sep 2025

- Developed in-context learning with large audio language models for SOTA in-the-wild speech deepfake detection (+54.9% avg Macro F1 improvement) (**NLP**, **RAG**, **ICL**, **Audio LLMs**, **model routing**).

Graduate Research Assistant | *Purdue University (Dr. Yung-Hsiang Lu)*, West Lafayette, IN Jan 2024 - Present

- Developed music tutoring models: 6× higher F1 (AAAI 2025) and 2× better missed-note detection with LadderSym (ICLR 2026) (**PyTorch**, **NLP**, **Audio Transformers**).
- Built generative music-mistake pipelines on HPC and Linux, producing two datasets.
- Organized 'AI for Music' workshops at AAAI 2025, IEEE ICME 2025, and NeurIPS 2025 (**leadership**, **communication**).

Research Assistant | *National Cheng Kung University (Dr. Hsun-Ping Hsieh)*, Taiwan Mar 2023 - Aug 2023

- Raised cancer-screening invitation accuracy to 83% with scikit-learn tree classifiers (**ML**, **statistics**, **data science**).
- Improved flash-chip yield by 17% with ensembling and Gaussian processes (**NumPy**, **pandas**).

Projects

Code Understanding with GitHub Issues and RAG | (**ChatGPT API**, **REST**, **Git**)

- Built a RAG-based doc tool with GitHub issues; raised README clarity by 116% and code clarity by 16%.

Multi-Agent Atari Self-Play | (**OpenAI Gym**, **PettingZoo**, **PyTorch**, **RL**, **Multi-agent systems**)

- Used DQN/PPO self-play to improve Atari win rates by 11–32%.

Spectral Image Inpainting | (**MATLAB**, **C++**, **CUDA**)

- Recovered 98% corrupted hyperspectral data at 0.02 RMSE using convex optimization and deep learning.

Honors & Awards

- Outstanding Student Scholarship: top 2 students in the NCKU-Purdue program (2020 - 2023).
- AAAI Student Scholarship (2025), selected for NeurIPS volunteer (2025).

Publications

- **B. Chou**, P. Jajal, N. J. Eliopoulos, J. C. Davis, G. K. Thiruvathukal, K. Y.-J. Yun, H.-W. Dong, Y.-H. Lu, "[On a Separate Note: Robust Score-Informed Note Separation with a Two-Stream TFC-TDF U-Net and Adaptive Set Ownership](#)", Submitted to *ICASSP* 2027
- **B. Chou**, Y. Zhu, S. Koppiseti, "[ICLAD: In-Context Learning with Comparison-Guidance for Audio Deepfake Detection](#)", *ACL* 2026 Findings
- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y. Lu, "[AdaPerceiver: Transformers with Adaptive Width, Depth, and Tokens](#)", *CVPR* 2026 Findings
- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y. Lu, "[Inference-Time Alignment of Diffusion Models with Evolutionary Algorithms](#)", *CVPR* 2026 Findings
- **B. Chou**, P. Jajal, N. J. Eliopoulos, J. C. Davis, G. K. Thiravathukal, K. Y.-J. Yun, Y.-H. Lu, "[LadderSym: A Multimodal Interleaved Transformer for Music Practice Error Detection](#)", *ICLR* 2026 (top 3.85%)
- **B. Chou**, P. Jajal, N. J. Eliopoulos, T. Nadolsky, C.-Y. Yang, N. Ravi, J. C. Davis, K. Y.-J. Yun, Y. Lu, "[Detecting Music Performance Errors with Transformers](#)", *AAAI* 2025
- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y.-H. Lu, "[Token Turing Machines are Efficient Vision Models](#)", *WACV* 2025

Patent Pending

- System and Method for Detecting Musical Performance Errors, Patent Application 2025.
- ICLAD: In-Context Learning with Comparison-Guidance for Audio Deepfake Detection, Provisional Patent Application 2025.